

Chapter 1

QUBIC: A Bolometric Interferometer (BI)

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Introduction

Observations of the Cosmic Microwave Background (CMB) have become one of the most powerful probes in modern cosmology, providing stringent constraints on the composition, geometry, and evolution of the Universe. In particular, the polarisation of the CMB carries unique information about the physics of the early Universe. Of special interest is the curl-like component of the polarisation field, known as B -modes, which can be generated by primordial tensor perturbations produced during Inflation. The detection of such primordial B -modes would provide compelling evidence for the quantum origin of cosmological fluctuations and offer a direct probe of the energy scale of Inflation. However, measuring B -mode polarisation remains extremely challenging. The expected signal is several orders of magnitude weaker than the temperature anisotropies and is significantly contaminated by astrophysical foregrounds,

such as Galactic dust and synchrotron emission, as well as by instrumental systematics. Achieving the required sensitivity and control over these systematics therefore necessitates innovative experimental strategies.

The Q & U Bolometric Interferometer for Cosmology (QUBIC¹) [1] has been designed to address these challenges by combining the high sensitivity of bolometric detectors with the intrinsic systematic control of interferometric techniques. Unlike traditional imaging experiments, QUBIC directly measures the interference pattern produced by an array of horns. This approach provides multiple independent constraints on the instrument response, which can be exploited through a self-calibration procedure to mitigate instrumental systematics, discussed further in Chapter 3. In addition, the interferometric nature of the instrument produces a frequency-dependent synthesized beam, which encodes spectral information directly in the measured data. This property enables the spectral imaging capabilities of QUBIC and allows a significant improvement in map-making and component-separation performance, as discussed in Chapter 4.

In this chapter, we present the QUBIC experiment in detail. We first describe the instrumental concept and its key components, with particular emphasis on the principles of bolometric interferometry. We then discuss the scientific objectives of QUBIC. Finally, we present the main instrumental subsystems of the telescope, along with their respective roles and limitations.

The QUBIC collaboration published a series of eight papers in 2022 describing the instrument in detail. These articles cover the overall project [1], its unique spectral capabilities [2], and comprehensive descriptions of its key instrumental components: instrument laboratory characterisation [3], performance of detectors and electronic readout [4], cryogenic system [5], Half Wave Plate rotator system [6], back-to-back horns array [7] and the optical design and specific beam [8]. They form the foundation of this chapter.

1.1 Telescopes for CMB physics

1.1.1 Classical approach: Imager

Before introducing QUBIC instrument, it is interesting to first present the design adopted by most modern CMB experiments, called *direct imaging*. Imagers are long-exposure telescopes directly forming images of the sky on their focal planes, where detectors are located. Imagers directly measure the sky intensity projected onto the focal plane, producing sky maps after scanning and reconstruction. This design is widely adopted by modern CMB experiments such as Planck [9], BICEP2/Keck Array [10], Simons Observatory [11] or LiteBIRD [12].

The angular resolution is fundamentally limited by diffraction, and scales as :

$$\theta \sim \frac{\lambda}{D}, \quad (1.1)$$

where λ is the observing wavelength and D the telescope diameter. As a consequence, large aperture telescopes are required to observe small angular scales.

¹<https://www.qubic.org.ar/>

These experiments use thousands of detectors, each observing a small patch of the sky independently. These detectors measure the total incoming power from photons. CMB polarisation observations require placing detectors after a polariser, or using specific polarisation-sensitive detectors, that need to be extremely sensitive detectors, operating close to the photon noise limit. The photon noise is a fundamental statistical noise, intrinsically limiting detectors' sensitivity. It consists of two terms : *shot noise* coming from the Poisson fluctuations of the discrete arrival of photons, scaling proportionally to frequency and photon flux, and *wave noise* due to their boson nature, photons arrive at the detector in correlated bunches. The dominant detector technology used in modern CMB instruments is the bolometer, which currently equips nearly all major experiments.² A bolometer measures the power of incoming radiation through the temperature increase it induces in an absorber thermally coupled to a cold bath. The absorbed power raises the temperature of a suspended thermal mass, whose temperature is monitored by a sensitive thermometer. In modern CMB experiments, this thermometer is typically a Transition Edge Sensor (TES), whose resistance varies sharply with temperature. By measuring this resistance variation, the incident optical power can be reconstructed.

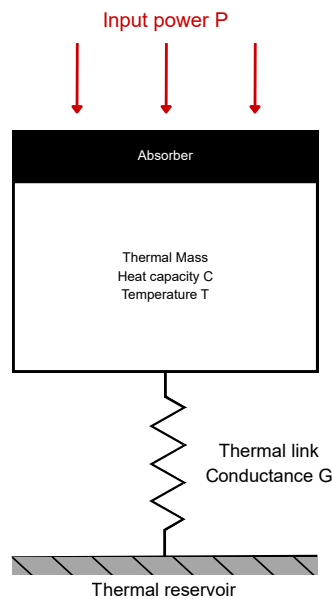


Figure 1.1: Diagram of a bolometer. Incoming radiation is absorbed by an absorber thermally coupled to a suspended thermal mass. The resulting temperature increase is measured by a thermometer, allowing the incident optical power to be estimated.

As detectors observe different regions of the sky independently, the ideal white-noise sensitivity improves with the number of detectors. Assuming uncorrelated white noise, the noise RMS of one detector scales as :

$$\sigma \propto \frac{1}{\sqrt{t}}, \quad (1.2)$$

²Another type of detector, known as Kinetic Inductance Detectors (KIDs), is the subject of extensive R&D efforts for future CMB experiments [13, 14]. KIDs are already employed in instruments such as NIKA2 [15] and TolTEC [16].

where t is the observing time. For N_{det} independent detectors observing simultaneously, the sensitivity therefore scales as :

$$\sigma \propto \frac{1}{\sqrt{N_{\text{det}} t_{\text{obs}}}}. \quad (1.3)$$

As a consequence, modern CMB experiments aim at maximizing the number of detectors populating the focal plane, representing a large fraction of the instrument complexity and cost. The number of detectors has dramatically increased over the last decade³.

1.1.2 Innovative design: Bolometric Interferometry

Interferometry was the dominant design for several CMB experiments in the early 2000s. Important examples include VSA [17], DASI [18] or CBI [19]. Interferometers can be separated into two main categories: *Coherent Interferometry* (CI), sometimes called *Heterodyne Interferometry*, and *Bolometric Interferometry* (BI).

In coherent interferometry, the instrument measures correlations between antennas, similarly to classical radio interferometry. Each antenna receives an electric field $E_i(t)$, which is amplified and pairwise correlated with the signals from other antennas through complex correlators. The measured quantity is called the *visibility*:

$$V(u, v) = \langle E_i E_j^* \rangle, \quad (1.4)$$

which corresponds to a Fourier mode of the sky, depending on the distance between the two antennas. One major advantage of this approach is its excellent control of systematic effects, due to the comparison between baselines. Interferometers also have access to Fourier modes of the sky, directly related to the power spectrum. In addition, their optical design is generally simpler than large imaging telescopes. However, coherent interferometers suffer from several major limitations. First, the correlators need amplifiers, whose noise complicates high-sensitivity measurements at CMB frequencies. The correlators also suffer from a very limited bandwidth: many are needed to observe enough photons. More importantly, the number of correlators scales as :

$$N_{\text{corr}} \sim N_{\text{ant}}^2, \quad (1.5)$$

making large interferometric arrays extremely expensive and technically complex. This scalability problem is the main reason why direct imagers progressively replaced coherent interferometers in modern CMB experiments⁴.

Bolometric interferometry aims at combining the advantages of interferometry with the high sensitivity of bolometric detectors. In BIs, electronic correlators are replaced by an optical combiner, naturally producing interference patterns from many baselines simultaneously. Amplifiers are removed, and the signal is directly measured using

³BICEP2/Keck Array (2015): ~ 3000 detectors, Simons Observatory (2025): $\sim 60\,000$ detectors.

⁴According to a presentation by Dan Werthimer in 2010 [20], the cost *per GHz of bandwidth* can be approximated by $1200N + 5N^2$, where N is the number of antennas. For the QUBIC Technical Demonstrator ($N = 64$) with a bandwidth of approximately 40 GHz, this corresponds to a cost of about 4 million USD (2010 value). Extrapolating to the QUBIC Full Instrument ($N = 400$) yields an estimated cost of approximately 50 million USD.

high-sensitivity bolometers. In bolometric interferometers, the different baselines are produced using an array of horns, which can be seen as an array of Young’s holes [21]. The signals are then combined through an optical system composed of large mirrors, focusing the resulting interference pattern onto a focal plane where detectors are located. Bolometric interferometry therefore aims at mitigating the scalability limitations of coherent interferometry while preserving its strong control of systematic effects. In addition, bolometers provide significantly higher sensitivity than coherent amplifiers at CMB frequencies ([100–300] GHz). Nevertheless, several important challenges remain. Bolometric interferometers rely on complex optical systems involving many elements (large horn arrays, optical combiners, switches, etc.), increasing both instrumental complexity and the number of potential systematic effects. Moreover, the synthesized beam produced by the instrument is significantly more complex than that of direct imagers and requires extremely precise characterisation for high-precision cosmological measurements. On the other hand, the large number of horns provides a high number of baselines, enabling self-calibration techniques that offer exceptional control of systematic effects and constitute one of the main scientific advantages of QUBIC.

The QUBIC experiment was designed to explore the possibility of constraining B-modes with Bolometric Interferometry, exploiting its 2 unique features: self-calibration and spectral imaging.

1.2 QUBIC Experiment

1.2.1 QUBIC Collaboration

During the 2000s, multiple projects started to develop around the concept of Bolometric Interferometry. We can cite Millimetre-wave Bolometric Interferometer (MBI) [22, 23], Einstein Polarization Interferometer for Cosmology (EPIC) [24], Background Radiation INterferometer (BRAIN) [25] or CMBPol [26]. The different members of these collaborations joined together to work on a common project : the Q & U Bolometric Interferometer for Cosmology⁵ (QUBIC) [27, 28, 29, 30, 1]. QUBIC is an international collaboration, with participations from France, Italy, Argentina, Ireland, the USA, the UK, and more recently Morocco.

The fundamental concept of QUBIC as a Bolometric Interferometer was first proposed in 2008 [27], culminating in the integration and validation of the Technical Demonstrator (TD) at APC⁶ in Paris, between 2018 and 2020. The Technical Demonstrator (TD) of the instrument was sent to Argentina in 2021. The pre-installation checkout of the system in Salta, the largest city close to the observing site, started the same year. The installation on the observing site and commissioning started in 2022. For financial and political reasons, commissioning stopped in 2023. It started again in late 2025. At the time of writing, commissioning and validation activities are ongoing, QUBIC’s commissioning and validation continue. Figure 1.3 presents one of the first observations of the Moon obtained with QUBIC at the observing site in 2026.

⁵<https://www.qubic.org.ar/>

⁶<https://apc.u-paris.fr/>



Figure 1.2: QUBIC observing site and collaboration laboratories across the world.

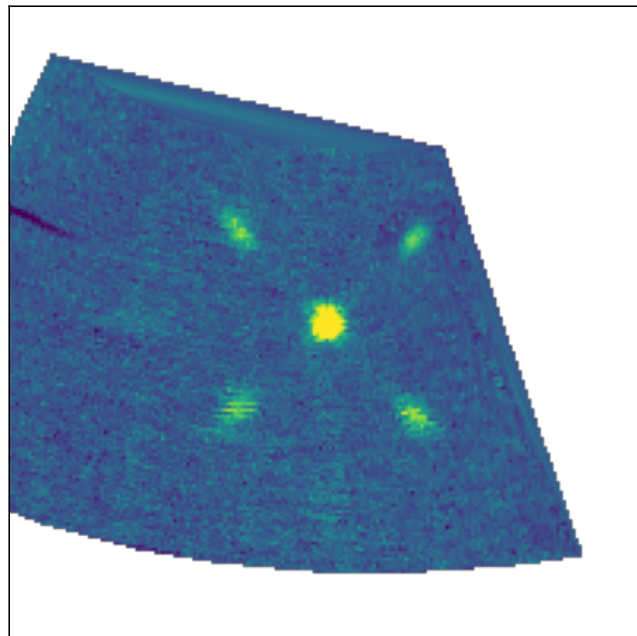


Figure 1.3: Observation of the moon at observing site in 2026. Map by Alexandre Huchet, from [31].

1.2.2 Observing Site

An observing site dedicated to CMB polarisation measurements must satisfy two main constraints. First, like any astronomical observatory, it must be located far from major sources of light pollution. Second, it must be situated at high altitude and in an extremely dry region. These latter requirements are intended to minimise atmospheric water vapour, which constitutes one of the main sources of noise⁷ for ground-based observations (see 1.4.5 and 11), since the water vapour density decreases rapidly with altitude. Only a few privileged observing sites were identified worldwide, as illustrated in Figure 1.4: the high-altitude regions of the Andes between Chile and Argentina, the Tibetan Plateau, Greenland, and the South Pole. From an atmospheric perspective, the South Pole, where experiments such as BICEP2 and Keck Array are located, provides exceptionally favourable observing conditions thanks to its extremely low PWV and highly stable atmosphere, despite its relatively modest altitude. However, the logistical challenges associated with operating in Antarctica (transportation, power supply, extremely low temperatures, limited repair capabilities, and restricted human presence) reduce the attractiveness of the site. Political and environmental considerations, governed by the Antarctic Treaty System, complicate the deployment of future facilities.⁸ In addition, only a limited fraction of the sky is observable from the South Pole, making large sky surveys impossible from this location. On the other hand, the same sky patch remains continuously visible, enabling uninterrupted long-duration observations, unlike at lower latitudes where targets rise and set during the day.

The most privileged observing site is the Atacama Desert in the Chilean Andes, where experiments such as Atacama Cosmology Telescope (ACT), Simons Observatory (SO), POLARBEAR, Simons Array, CLASS, and QUIET (as well as several non-CMB facilities) are located. This region is among the driest on Earth outside Antarctica, providing excellent atmospheric conditions for millimetre and sub-millimetre observations. Chile also has a long-standing history of astronomical collaboration, which has enabled the development of robust infrastructure and stable logistical support in this region [33]. On the Argentine side of the Andes, the Alto Chorrillos region⁹ hosts experiments such as QUBIC, as shown in Figure 1.5, and the LLAMA project. This site shares very similar atmospheric characteristics with the Atacama plateau, including high altitude and low precipitable water vapour. However, differences in long-term investment in astronomical infrastructure and the maturity of large-scale observatory ecosystems result in a less developed operational environment compared to northern Chile. In the Northern Hemisphere, Greenland and Tibet are also promising high-altitude or high-latitude sites for millimetre astronomy [34]. Greenland suffers from similar logistical limitations to the South Pole, particularly in terms of infrastructure and accessibility, despite excellent atmospheric conditions. Tibet, while offering high-

⁷The total optical power received by a detector can be written as the sum of the contributions from the sky and the atmosphere, $P_{\text{det}} = P_{\text{sky}} + P_{\text{atm}}$. Since the atmospheric loading dominates the sky signal, $P_{\text{atm}} \gg P_{\text{sky}}$, the associated photon noise is also dominated by the atmosphere, leading approximately to $\sigma_{\text{det}} \propto \sqrt{P_{\text{det}}} \simeq \sqrt{P_{\text{atm}}}$.

⁸For example, discussions surrounding the implementation of next-generation projects such as CMB-S4 have highlighted the increasing logistical and environmental constraints associated with Antarctic deployment.

⁹The Atacama Desert and Alto Chorrillos are separated by approximately 300km.

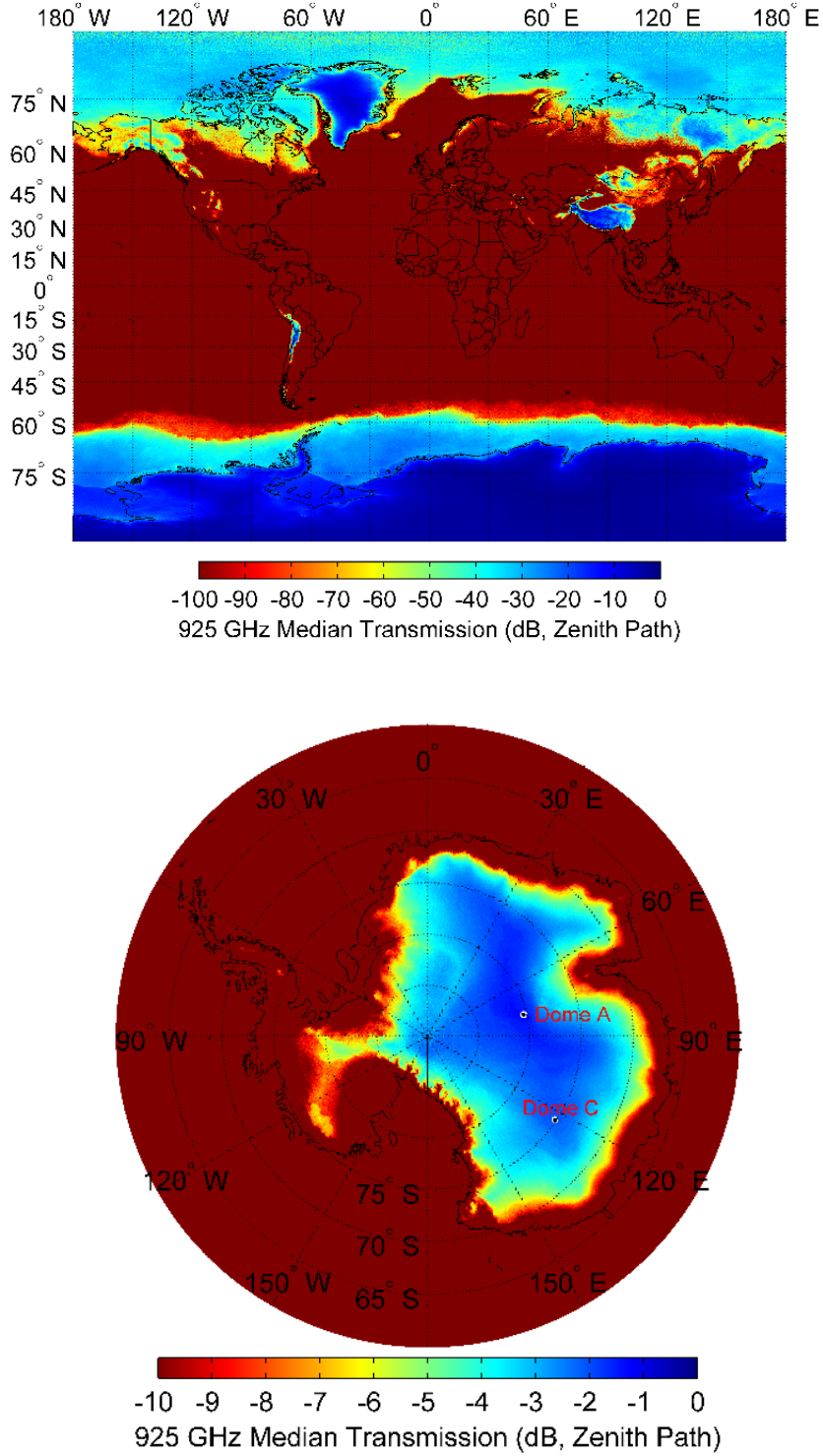


Figure 1.4: Median transmission at 925 GHz at Zenith across the world (top) and focus on the South Pole (bottom). The transmission is directly related to Precipitable Water Vapour (PWV) of the atmosphere in this region. The frequency of this map is not relevant for CMB observations, but it has been chosen as it amplifies the effect of atmosphere (whose transmission decrease with frequency, see Figure 11.6) for visibility purpose. For astrophysical observations, we are looking for the highest transmission, meaning the bluest regions on the maps. Taken from [32].

altitude dry sites, faces challenges related to infrastructure development and long-term operational and political stability, which currently limit its use for large-scale CMB observatories.



Figure 1.5: Picture of QUBIC during its inauguration in 2022, in Alto Chorrillos, Salta, Argentina.

1.2.3 Observation Sky Patch

For scientific surveys, the observed region on the sky matters. Experiments observing a large fraction of the sky are generally interested in large scale effects or high statistics on the data. On the opposite, some collaborations are observing only a small fraction of the sky, which is done to achieve higher signal-to-noise ratio. QUBIC is scheduled to be in the second case, observing a $\sim 15^\circ$ patch located at $(RA = 0^\circ, DEC = -57^\circ)$ during 3 years, as explained in [1]. The main advantage of this sky patch is the low contamination from dust and synchrotron [35], as well as being visible from QUBIC observing site about 40% of the time. Figure 1.6 shows QUBIC's target patch.

1.2.4 Science Objectives

CMB Polarisation

As suggested by its name, the QUBIC instrument was specifically designed to measure CMB polarisation. Several instrumental features were developed for this purpose. They are briefly introduced here and discussed in detail throughout this thesis.

The detection of primordial CMB B-modes requires both very high sensitivity and excellent control of instrumental systematic effects. QUBIC addresses the first requirement through the use of Transition Edge Sensors (TESs) [4], a class of bolometric detectors described in Subsection 1.3.6. To meet the second requirement, QUBIC employs *self-calibration*, a technique widely used in radio interferometry that consists in comparing measurements from different baselines in order to constrain instrumental

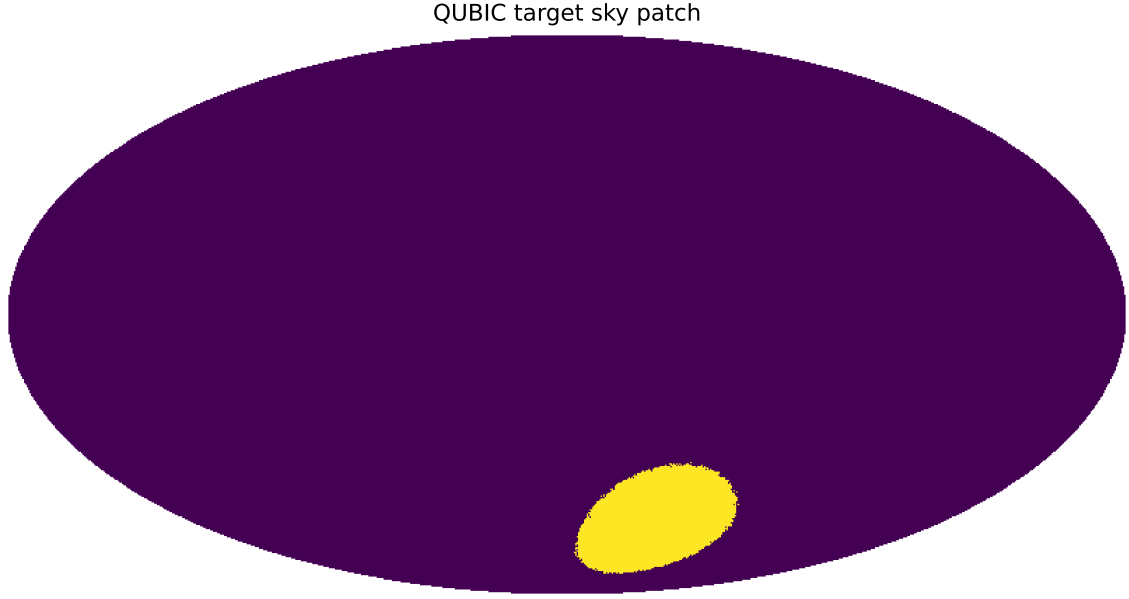


Figure 1.6: QUBIC patch on a sky map.

systematic effects [36, 37, 38, 39]. This method is discussed in Chapter 3, and more specifically in Subsection 3.2.1.

Another key feature of QUBIC is the use of a polarising grid coupled to full power detectors. This architecture strongly suppresses instrumental cross-polarisation effects [6]; further details are provided in Subsection 1.3.3.

Foreground Mitigation

Separating the CMB polarisation signal from astrophysical foreground contamination relies primarily on the different spectral dependences of the various emissions. In practice, this requires observations over multiple frequency channels in order to disentangle the different contributions.

For traditional imagers, this leads to a significant increase in instrumental complexity and cost, since dedicated detector arrays are generally required for each frequency band. Bolometric interferometers, however, naturally exhibit a frequency-dependent synthesized beam, allowing the spectral information to be spatially encoded across the focal plane. This unique feature, referred to as *spectral imaging*, is one of the main advantages of bolometric interferometry. It is presented in detail in Chapter 4, and constitutes the core of this thesis, as all chapters in Part III rely on this concept.

It is also important to note that an instrument designed to separate CMB emission from astrophysical foregrounds must necessarily be highly sensitive to the properties of these foreground emissions themselves. In this respect, QUBIC could also provide valuable measurements for Galactic science, even though this is not part of its primary scientific program.

Demonstrating Bolometric Interferometry

QUBIC is a unique instrument employing a technology that is entirely novel within the field of CMB observations. As such, it serves as a proof of concept for bolometric interferometry, demonstrating both its capabilities and its potential for future generations of CMB experiments.

1.3 QUBIC Instrument

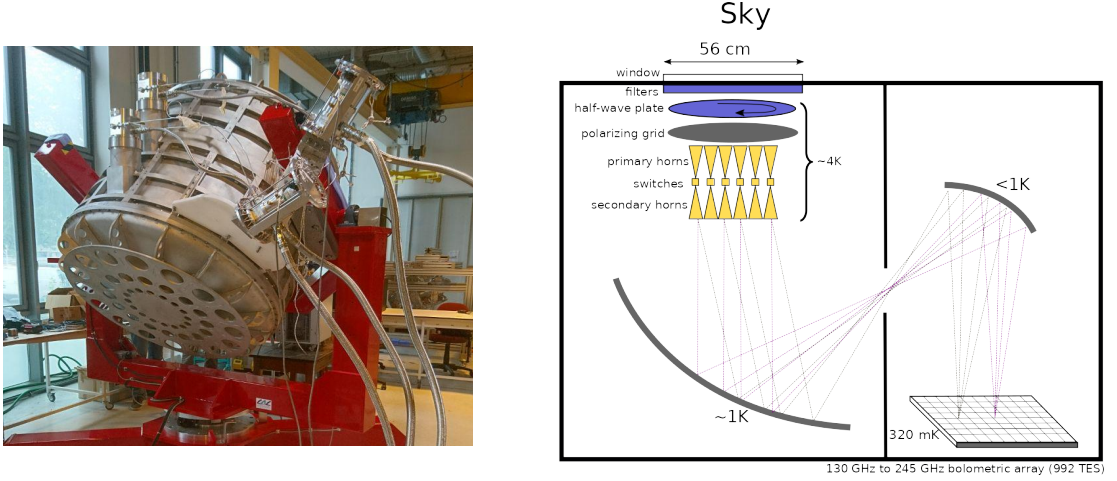


Figure 1.7: (left) Picture of QUBIC Instrument assembly hall at APC in Paris during integration phase. (right) Schematic of the instrumental design of QUBIC.

In this section, the full instrumental design of QUBIC is described. Before detailing the various optical and electronic components, we provide a brief overview of the overall design. Photons enter the instrument through a 56 cm transparent window, after which they are filtered to remove frequencies outside the observation bandwidth, thereby limiting unwanted thermal loading from high-energy radiation. The radiation then propagates into QUBIC’s cryostat [5]. The polarisation state of the incoming light is modulated by a rotating Half-Wave Plate (HWP), before passing through a polariser that selects a single polarisation component. The resulting signal is then distributed to an array of 400 back-to-back horns [7] located at the 4 K stage. The 400 split beams are subsequently reflected by two concave mirrors at 1 K, which focus the radiation onto the focal plane, where 992 Transition Edge Sensors (TESs) [4], cooled to approximately 320 mK, form an interference pattern. QUBIC observes the sky through the two atmospheric windows centred around 150 and 220 GHz.

1.3.1 Technical Demonstrator

In order to validate the instrumental concept of QUBIC, including bolometric interferometry, the cryogenic system, optics, detectors, and readout electronics, a Technical Demonstrator (TD) was constructed. The TD uses the same cryostat as the Full Instrument (FI), but with a reduced optical configuration: 64 horns instead of 400, a

smaller focal plane (one quarter of the full size), and smaller mirrors. The TD operates in a single frequency band at 150 *GHz*. Its primary objective is to validate the full end-to-end functionality of all QUBIC subsystems and to demonstrate the feasibility of bolometric interferometry under realistic observational conditions on the sky.

To transition from the TD to the FI, several components still need to be manufactured, tested, and installed. While the horn array, bolometers, and readout electronics have already been produced, the bolometers are still undergoing testing, and additional cabling for the readout electronics remains to be integrated. The mechanical switches for the horn configuration have not yet been manufactured, and additional optical components such as filters and the half-wave plate (HWP) have been ordered and are expected to be delivered within the year. This staging enables a relatively straightforward transition from the demonstrator phase to the full-scale instrument once on-site integration is completed.

1.3.2 Window & Filters

QUBIC observes the sky through a 56 cm diameter window, made with multi-layer high density polyethylene. This material is transparent to millimetre wavelengths, while being opaque to infrared signals to avoid thermal load in the instrument [40]. The aperture is followed by a series of filters which aims at cutting frequencies outside the observing windows allowed by atmosphere, between 130 and 245 GHz, with a notch filter to cut between 170 and 200 GHz to avoid the H_2O emission line (see Subsection 1.4.5). The TD filters only allow for observations between 130 and 170 GHz.

1.3.3 Half-Wave Plate & polariser

Right after the filters, two optical elements dealing with polarisation are placed: the Half-Wave Plate (HWP) [6] and the polariser. They are both cooled down to 4K to limit the thermal load of the instrument.

The HWP rotates in a step-by-step manner and modulates the sky signal depending on its rotation angle. After the HWP, the electric field

$$\vec{E} = \begin{pmatrix} E_x \\ E_y \end{pmatrix}$$

becomes:

$$\vec{E}' = \begin{pmatrix} E_x \cos(2\varphi(t)) + E_y \sin(2\varphi(t)) \\ E_x \sin(2\varphi(t)) - E_y \cos(2\varphi(t)) \end{pmatrix}, \quad (1.6)$$

where φ is the rotation angle of the HWP.

The motor is placed outside the cryostat, and its rotation is transmitted to the HWP via a glass fibre axis magnetically connected to a pulley system, rotating the plate with a belt (see Figure 1.8).

After the HWP, the signal enters a polariser, which transmits only the component aligned with its axis, effectively selecting a single linear polarisation state. The field becomes:

$$\vec{E}' = \begin{pmatrix} E_x \cos(2\varphi(t)) + E_y \sin(2\varphi(t)) \\ 0 \end{pmatrix}. \quad (1.7)$$

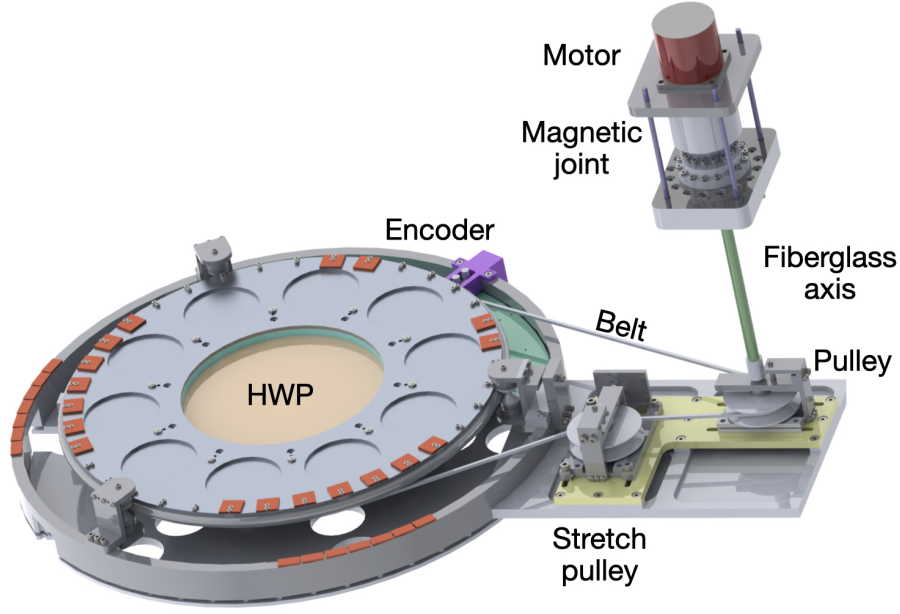


Figure 1.8: 3D model of the Half-Wave Plate rotator mechanism. Taken from [6].

The resulting detected signal mixes intensity and polarisation, allowing reconstruction with intensity-only detectors from trigonometric formula:

$$\begin{aligned}
 S &= |\vec{E}|^2 \\
 &= E_x^2 \cos^2(2\varphi) + E_y^2 \sin^2(2\varphi) + 2E_x E_y \cos(2\varphi) \sin(2\varphi) \\
 &= E_x^2 \frac{(1 + \cos(4\varphi))}{2} + E_y^2 \frac{(1 - \cos(4\varphi))}{2} + E_x E_y \sin(4\varphi) \\
 &= \frac{1}{2} (E_x^2 + E_y^2) + \frac{1}{2} \cos(4\varphi) (E_x^2 - E_y^2) + E_x E_y \sin(4\varphi) \\
 S &= \frac{1}{2} (I + Q \cos(4\varphi) + U \sin(4\varphi)) .
 \end{aligned} \tag{1.8}$$

The fixed 7 positions for the HWP, separated by 15° [6], and associated signals are summarised in Table 1.1.

Position	0	1	2	3	4	5	6
φ [rad]	0	$\pi/12$	$\pi/6$	$\pi/4$	$\pi/3$	$5\pi/12$	$\pi/2$
S	$I + Q$	$I + \frac{Q+\sqrt{3}U}{2}$	$I - \frac{Q-\sqrt{3}U}{2}$	$I - Q$	$I - \frac{Q+\sqrt{3}U}{2}$	$I + \frac{Q-\sqrt{3}U}{2}$	$I + Q$

Table 1.1: Intensity measured as a function of I , Q , and U for the seven HWP steps of QUBIC.

Although it may seem wasteful to discard half of the incoming photons with the polariser, this is a key instrumental choice. By selecting a single polarisation state early in the optical chain and using intensity-only detectors, the instrument becomes significantly less sensitive to internal cross-polarisation effects arising from optical components such as horns and mirrors, as well as from reflections within the instrument

downstream of the polariser. Since mitigating such systematics would otherwise require substantial effort, it is preferable to sacrifice part of the incoming signal in exchange for improved control of the instrument’s polarisation response. The effectiveness of this design has been demonstrated in [3], where the instrumental cross-polarisation was measured to be below 0.4% at the 95% confidence level. This value is significantly lower than the percent-level cross-polarisation typically reported for many classical CMB imagers, highlighting one of the main advantages of the QUBIC optical architecture.

1.3.4 Horns Array

The next and key part of the interferometric design of QUBIC is the horns array [7]. It consists of 400 back-to-back corrugated horns at 4 K, separated by 14 mm. Corrugations, periodic grooves machined inside the horn walls meant to improve beam symmetry and reduce cross-polarisation systematics, are shown in Figure 1.9. Both back-to-back horns are designed to be identical, to observe the sky and illuminate the optical combiner with a Full-Width at Half-Maximum (FWHM) of 12.9° at 150 GHz. Between them is a mechanical switch, thought to remotely close or open each horn in the array separately, a mandatory feature to perform self-calibration [39] (see 3.2.1).

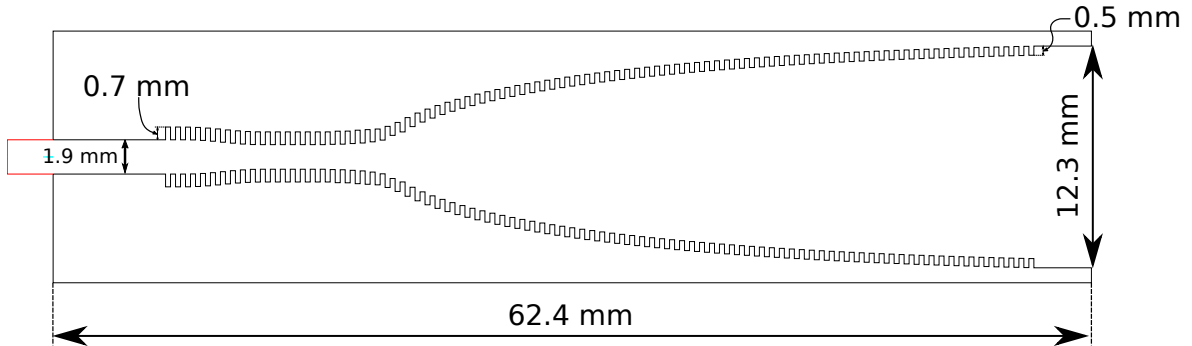


Figure 1.9: Schematic of the profile of the corrugated horn. Taken from [7].

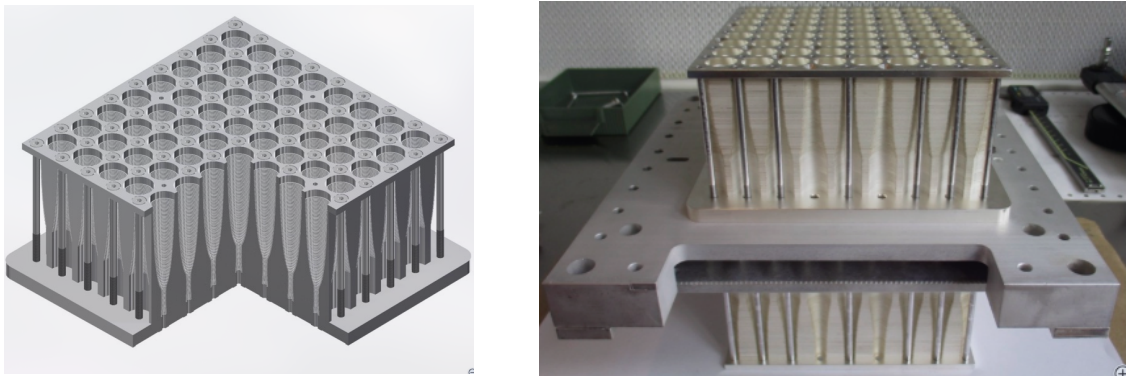


Figure 1.10: (left) 3D model of horn array. (right) Picture of the fully integrated back-to-back horn array. Taken from [7].

1.3.5 Optical Combiner

The multiple beams produced by the horn arrays then illuminate the optical combiner [8], two concave mirrors, to focus light into the focal plane, producing an interference pattern. The two mirrors have a diameter of 600 mm and the system has a focal length of 30 cm. They are cooled down at 1 K.

1.3.6 Bolometers & readout electronics

The focal plane of QUBIC consists of an array of bolometric detectors that measure incoming radiation through the temperature rise induced by photon absorption. In particular, QUBIC employs Transition Edge Sensors (TES) [4], chosen for their exceptional sensitivity when operated within the superconducting transition. TESs are based on a superconducting NbSi thin film thermally coupled to a heat bath. The detectors are voltage-biased to measure the resistance variations and operated near their critical temperature, typically around $T_c \simeq 300$ mK. In this regime, small variations in temperature induced by absorbed optical power produce large variations in resistance, as illustrated in Figure 1.11. Below the critical temperature, the device is in a superconducting state with negligible resistance. The superconducting transition occurs over a narrow temperature range (approximately 0.47–0.49 K for the sample shown, used during R&D process, not in QUBIC instrument¹⁰), where the resistance exhibits a strong dependence on temperature. This steep slope is essential for achieving high responsivity. Under constant voltage bias, TES detectors operate with strong electro-thermal feedback. The electrical power dissipated in the TES is given by the Joule effect, $P_{\text{Joule}} = \frac{V^2}{R}$, where V is the bias voltage and R the TES resistance. As temperature rises by heating, the resistance strongly increases, leading to lower heating through Joule effect. This negative feedback loop, called *electro-thermal feedback*, stabilises the TES temperature within the superconducting transition and reduces the effective thermal time constant.

Variations in TES resistance are measured using a Superconducting QUantum Interference Device (SQUID). The SQUID senses the corresponding change in current as a variation in magnetic flux and converts it into an output voltage. Each quarter of the focal plane, consisting of 248 TESs, is connected to two blocks of 128 SQUIDS operating at 1 K. The output signals from the SQUID array are then read out by an Application Specific Integrated Circuit (ASIC) operating at 40 K. To reduce the number of readout cables, and consequently the thermal load and system complexity, a multiplexing scheme is implemented to transmit the detector information.

The detector performance is characterised by the Noise Equivalent Power (NEP), which quantifies the input power that produces a signal equal to the detector noise in a 1 Hz bandwidth. Laboratory measurements report an NEP of 4.7×10^{-17} W/ $\sqrt{\text{Hz}}$, while measurements using the TD readout chain yield 2.05×10^{-16} W/ $\sqrt{\text{Hz}}$. This increase is attributed mainly to aliasing effects in the readout chain. This limitation is expected to be mitigated in the final instrument by implementing Nyquist inductance

¹⁰This measurement comes from the TES sample P41, while the one currently used for the TD is the sample P87, with different properties (in particular, a different critical temperature). It is used only for illustrative purposes, associated with the resistance-versus-temperature curve.

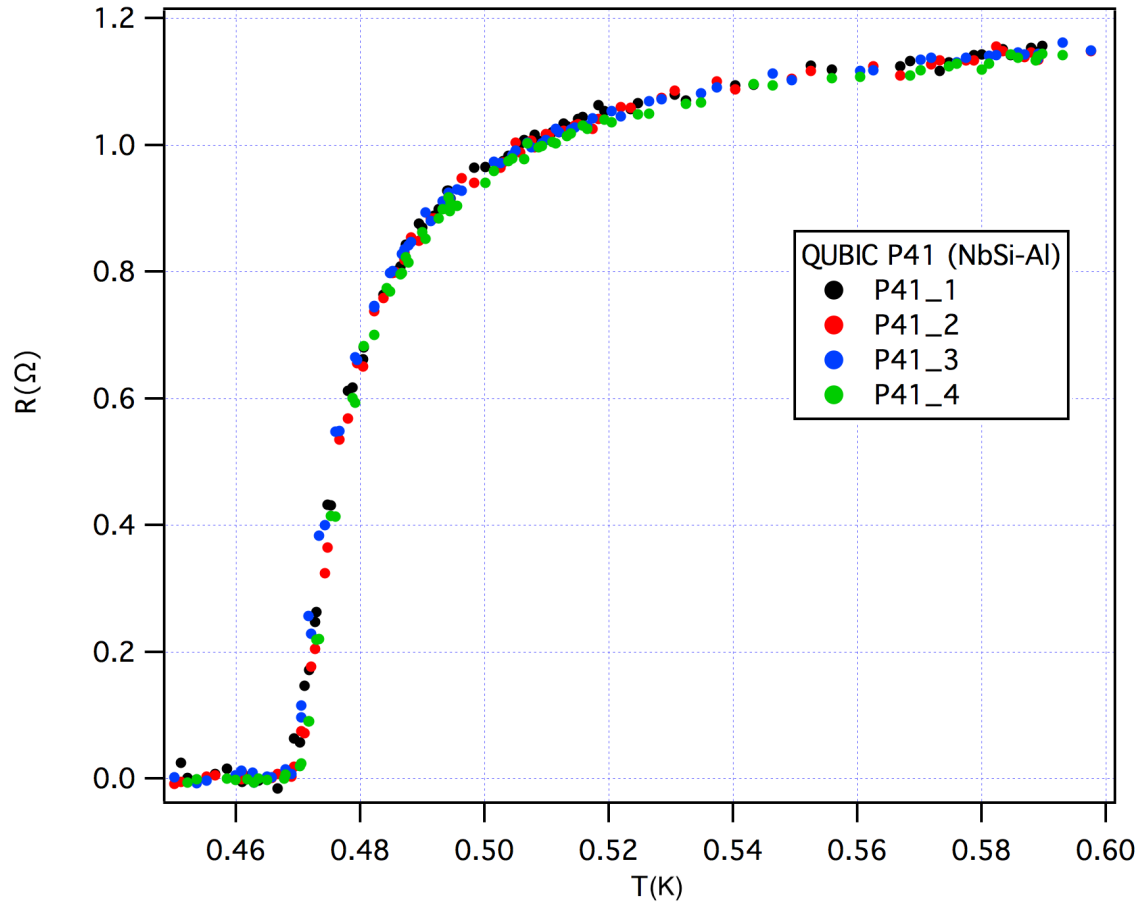


Figure 1.11: Resistance of four TESs as a function of temperature. The sharp superconducting transition provides high sensitivity to small temperature variations. Taken from [4].

filtering, in order to reach background-limited performance, where detector sensitivity is dominated by photon noise rather than readout noise. [4].

1.3.7 Cryogenic system

The cryogenic system, described in [5], is a key element enabling high-precision measurements. By cooling the different components of the detection chain, it reduces the thermal load of the instrument while ensuring that the bolometers operate within their superconducting transition. Several cooling techniques are employed to reach the different temperature stages. The system operates under high-vacuum conditions, which is essential for minimising thermal conduction and enabling efficient thermal isolation between stages. The filters, HWP, polariser, and horn array are cooled down to 4 K using a pulse-tube cryocooler based on the compression and expansion cycle of helium gas. The two mirrors are then cooled further to approximately 1 K. This stage is achieved using sorption coolers. A sorption cooler operates by condensing helium gas into a liquid state, which accumulates in the evaporator, reaching an equilibrium between liquid and vapour phases. An activated charcoal cryopump then adsorbs the helium vapour, breaking this equilibrium and driving endothermic evaporation of the liquid helium, which leads to a reduction in temperature. Once all liquid helium has evaporated, the cooler is thermally isolated from the instrument. The helium is then released from the cryopump by heating it with an electrical resistor, allowing the cycle to restart. Finally, cooling of the detector stage to approximately 320 mK is achieved using a two-stage sorption cooler based on ^3He and ^4He . Both stages operate on the same principle, with the first stage pre-cooling the second to reach the required sub-Kelvin temperatures.

1.4 Instrumental Designs

1.4.1 Legacy design : Dual Bands

The original QUBIC design, known as Dual Bands (DB), consisted of two focal planes, each composed of 992 detectors. One focal plane was dedicated to the 150 GHz channel, while the other targeted the 220 GHz channel. This configuration, shown in Figure 1.12, incorporated a dichroic filter. The dichroic acts as a prism which deflects light at an angle depending on the wavelength. It was designed to cleanly separate the two frequency bands at 150 and 220 GHz. This architecture offers the advantage of providing two uncorrelated frequency channels, a feature that can be further enhanced by the spectral imaging capabilities described in Chapter 4. However, the implementation of such a dichroic proved challenging: the required 90° beam deflection, combined with severe spatial constraints that would have required a non-symmetric dichroic geometry, made fabrication impractical. As a result, this design was eventually abandoned. Nevertheless, results based on this instrumental configuration are still presented throughout this thesis, as one of my initial tasks was to compare the sensitivity of the DB design with its alternative, the Ultra-Wide Band (UWB) design.

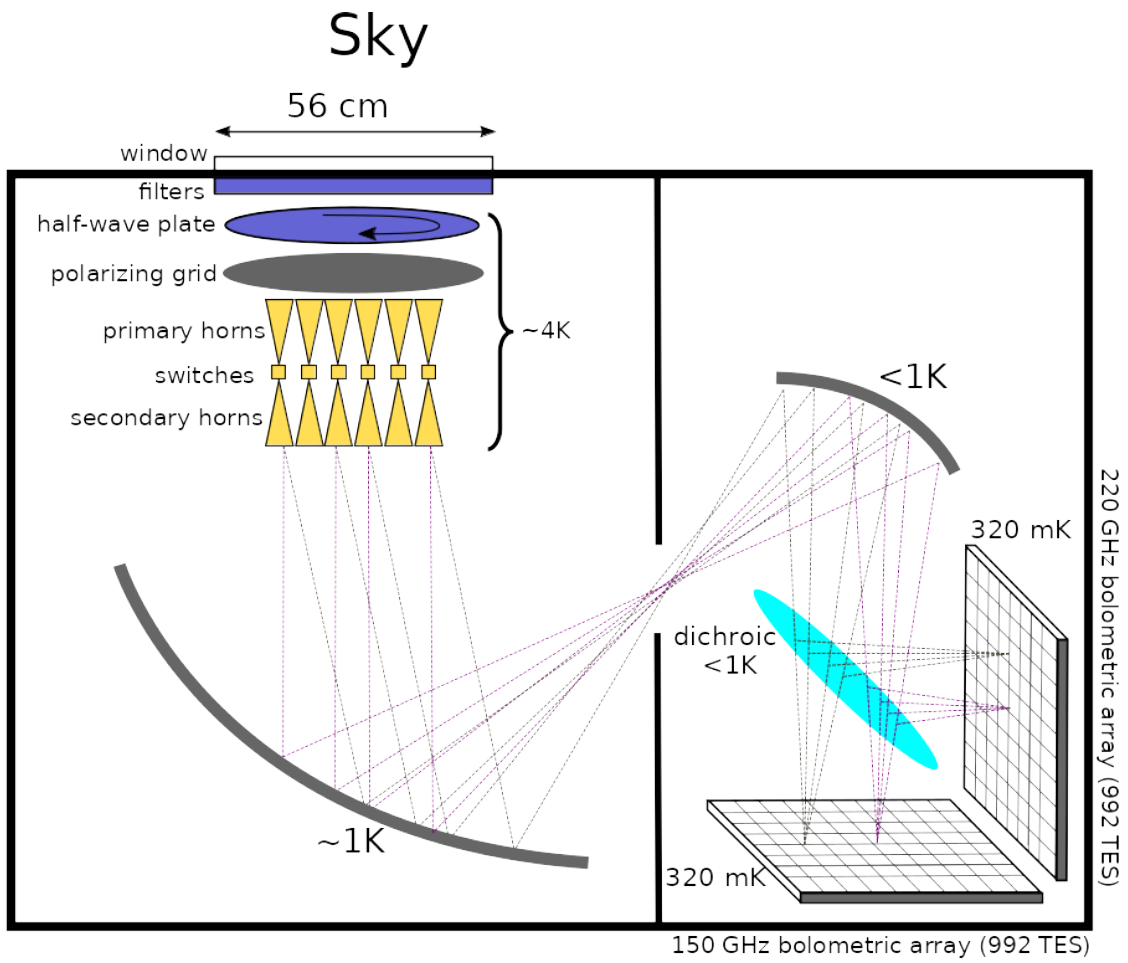


Figure 1.12: **Dual Bands design:** includes two focal planes, observing at 150 & 220 GHz respectively, separated by a dichroic filter.

1.4.2 Interferometry-oriented design : Ultra Wide Band

Fortunately, the spectral imaging capability of QUBIC provides an obvious solution to the problem of the dichroic. The solution is the Ultra-Wide Band (UWB) design, based on a single focal plane composed of 992 detectors, which simultaneously observe both the 150 GHz and 220 GHz frequency channels. The corresponding optical configuration is presented in Figure 1.13. In this approach, spectral imaging is used to separate the contributions from different frequency components by reconstructing sub-bands in post-processing. This removes the need for a dichroic filter, while also reducing the number of detectors by a factor of two, along with their associated cost. However, this design introduces several challenges. First, it relies entirely on the performance of the spectral imaging reconstruction, and therefore on the ability of the QUBIC collaboration to accurately disentangle the observed signal, which has its own limitations and requires dedicated development (see Chapter 4). Additionally, operating over a broader frequency band increases the thermal loading on the detector array, particularly due to high-frequency contributions dominated by atmospheric emission and astrophysical dust. The extent to which this impacts sensitivity depends on the resulting increase in photon noise and on the efficiency of the filtering strategy implemented at the instrument level.

Despite these limitations, the UWB design offers several important advantages. The main benefit is a significant simplification of the instrumental architecture: a single on-axis optical chain, no dichroic element, reduced cryogenic complexity due to fewer readout electronics and thermal connections, and a simpler mechanical structure. Moreover, the reduction in detector count also decreases the overall detector noise contribution. Reducing the detector count does not degrade sensitivity, and can even be beneficial through a reduction in detector noise, since the same frequency range is observed with fewer detectors. Nevertheless, this simplification comes at the cost of increased reliance on spectral imaging, which introduces its own limitations.

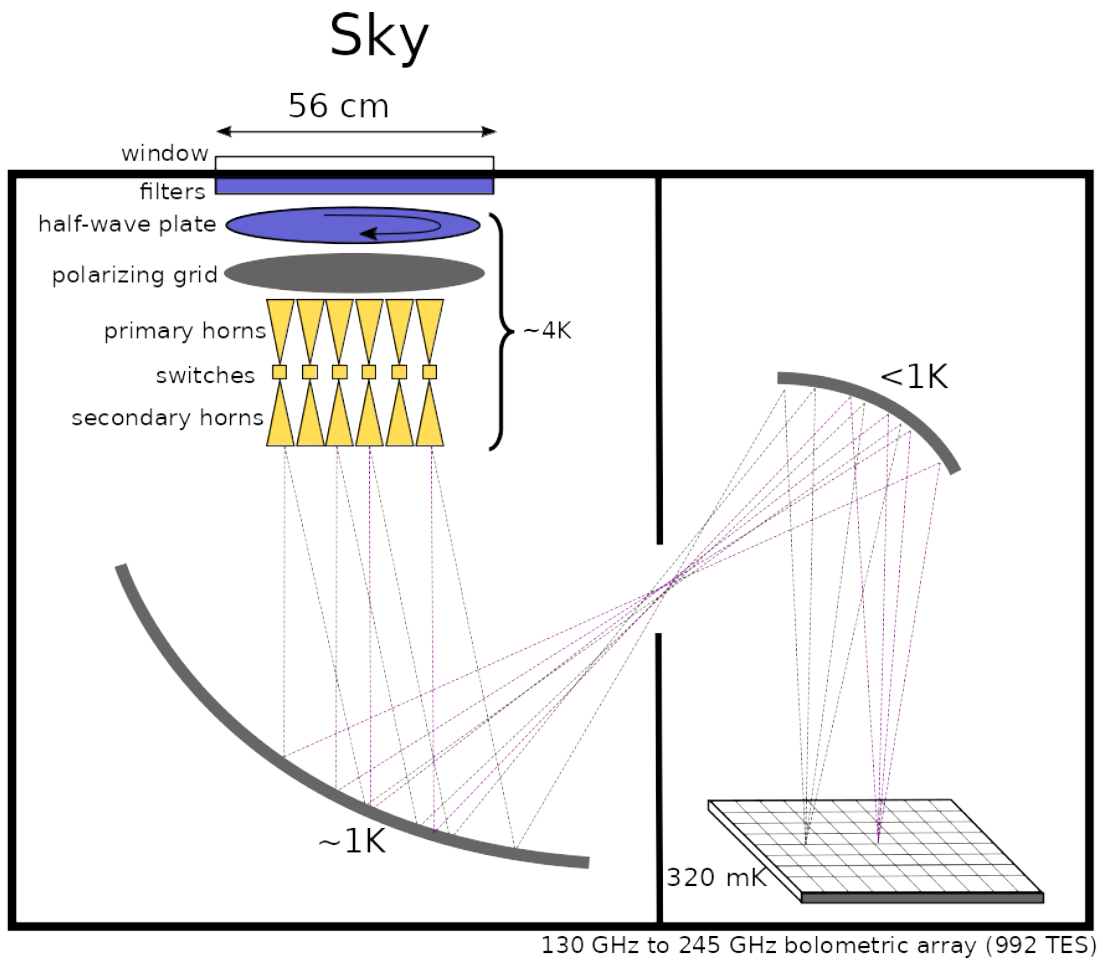


Figure 1.13: **Ultra Wide Band design:** includes only one focal plane, observing at both 150 & 220 GHz simultaneously.

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